

# JEE MAIN REVISION MASTER: MATRICES AND DETERMINANTS SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 368 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                            | Frequency | Category                      |
|---------------|---------------------------------------------------------|-----------|-------------------------------|
| Very High     | Properties of Determinants & System of Linear Equations | 250 PYQs  | Determinants & Linear Systems |
| Very High     | Matrix Inversion, Adjoint & Powers                      | 121 PYQs  | Matrix Algebra                |

## 2. Master Formula Reference Card

| Concept / Topic                                         | Mathematical Formulation & Governing Equations                                                                                                                                                       |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Properties of Determinants & System of Linear Equations | $\det(kA) = k^n \det(A)$ , $\text{Unique: } \Delta \neq 0$ , $\text{Infinite: } \Delta = \Delta_x = \Delta_y = \Delta_z = 0$ , $\text{No sol: } \Delta = 0 \text{ \& at least one } \Delta_i \neq 0$ |
| Matrix Inversion, Adjoint & Powers                      | $A \cdot \text{adj}(A) = (\det A) I_n$ , $\det(\text{adj}(A)) = (\det A)^{n-1}$ , $\text{adj}(\text{adj}(A)) = (\det A)^{n-2} A$                                                                     |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Properties of Determinants & System of Linear Equations (Very High Yield — 250 PYQs)

**Core Principle:** Evaluation of determinants, Cramer's rule, and consistency conditions for non-homogeneous and homogeneous linear systems.

■■ **Caution / JEE Pitfall:** For an  $n \times n$  matrix  $A$ ,  $\det(kA) = k^n \det(A)$ , NOT  $k \det(A)$ !

■ **Speed Trick:** A homogeneous system  $AX = 0$  has non-trivial (infinite) solutions if and only if  $\det(A) = 0$ .

**Representative Source PYQs:**

- [2026 Evening Shift] Q#1 (MCQ): "If the system of linear equations :  $x+y+z=6$   $x+2y+5z=10$   $2x+3y+\lambda z=\mu$  has infinitely many s..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "If the system of equations :  $x+y+z=5$   $x+2y+3z=9$   $x+3y+\lambda z=\mu$  has infinitely many solutions,..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "If the system of equations  $x+5y+6z=4$   $2x+3y+4z=7$   $x+6y+az=b$  has infinitely many solutions, then the point  $(a,b)$  lies on the line..." [Ans: N/A]

### 3.2 Matrix Inversion, Adjoint & Powers (Very High Yield — 121 PYQs)

**Core Principle:** Properties of adjoint and inverse matrices, matrix multiplication transpose  $(AB)^T = B^T A^T$ , and symmetric/skew-symmetric decomposition.

■ **Caution / JEE Pitfall:** A skew-symmetric matrix of ODD order has determinant equal to ZERO ( $\det(A) = 0$ ).

■ **Speed Trick:** For a  $3 \times 3$  matrix:  $\det(\text{adj}(A)) = (\det A)^2$ , and  $\det(\text{adj}(\text{adj}(A))) = (\det A)^4$ .

**Representative Source PYQs:**

- **[2026 Evening Shift] Q#1 (MCQ):** "If  $f: N \rightarrow Z$  is defined by  $f(n) = \begin{cases} n & \text{if } n \text{ is odd} \\ -5n & \text{if } n \text{ is even} \end{cases}$  then  $f(2k+1) + f(2k+2) + f(2k+3) + \dots + f(4k+1)$  is equal to \_\_\_\_\_" **[Ans: N/A]**
- **[2026 Evening Shift] Q#1 (MCQ):** "Let  $P = [p_{ij}]$  and  $Q = [q_{ij}]$  be two square matrices of order 3 such that  $q_{ij} = 2^{i+j-1} p_{ij}$  and  $\det(Q) = 2^{10}$ . Then the value of  $\det(P)$  is \_\_\_\_\_" **[Ans: N/A]**
- **[2026 Evening Shift] Q#1 (MCQ):** "For the matrices  $A = \begin{bmatrix} 3 & -4 & 1 \\ -1 & 1 & 1 \end{bmatrix}$  and  $B = \begin{bmatrix} -29 & 49 & -13 & 18 \end{bmatrix}$ , if  $A^2 + \alpha \text{adj}(\text{adj}(A)) + \beta \text{adj}(A) \text{adj}(\text{adj}(A)) = B$  for some  $\alpha, \beta \in \mathbb{R}$ . Then  $(\alpha - \beta)^2$  is equal to \_\_\_\_\_" **[Ans: N/A]**

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

Let  $A = \begin{bmatrix} 1 & 1 & -1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & -1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$  satisfy  $A^2 + \alpha \text{adj}(\text{adj}(A)) + \beta \text{adj}(A) \text{adj}(\text{adj}(A)) = \begin{bmatrix} 2 & -2 & 2 & -2 & 0 & -1 & 0 & 0 & -1 \end{bmatrix}$  for some  $\alpha, \beta \in \mathbb{R}$ . Then  $(\alpha - \beta)^2$  is equal to \_\_\_\_\_

**Correct Answer: 4**

**Solution Analysis:** We need to find  $\alpha$  and  $\beta$  from  $A^2 + \alpha \text{adj}(\text{adj}(A)) + \beta \text{adj}(A) \text{adj}(\text{adj}(A)) = \begin{bmatrix} 2 & -2 & 2 & -2 & 0 & -1 & 0 & 0 & -1 \end{bmatrix}$ .

Let us solve step by step. 1. Given matrix  $A = \begin{bmatrix} 1 & 1 & -1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & -1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$ . 2. Find  $\det(A)$  Since the third row is  $(0, 0, 1)$ , expand along the third row:  $\det(A) = 1 \cdot \begin{vmatrix} 1 & 1 & -1 & 1 & 0 & 1 \\ 1 & 1 & -1 & 1 & 0 & 1 \end{vmatrix} = (-1)(0) - (1)(1) = -1$ . So,  $\det(A) = -1 \neq 0$ . Hence  $A$  is invertible. 3. Use NCERT property of adjoint of adjoint For an  $n \times n$  invertible matrix,  $\text{adj}(\text{adj}(A)) = (\det A)^{n-2} A$ .

### Benchmark Problem 2: JEE Main 2026 (Online) 2nd April Evening Shift (Q#2)

Consider the matrices  $A = \begin{bmatrix} 2 & -2 & 4 & -2 \end{bmatrix}$  and  $B = \begin{bmatrix} 3 & 9 & 1 & 3 \end{bmatrix}$ . If matrices  $P$  and  $Q$  are such that  $PA = B$  and  $AQ = B$ , then the absolute value of the sum of the diagonal elements of  $2(P+Q)$  is \_\_\_\_\_.

**Correct Answer: 34**

**Solution Analysis:**  $A^{-1} = \frac{1}{4} \begin{bmatrix} -2 & 2 & -4 & 2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -1 & 1 & -2 & 1 \end{bmatrix}$ .  $\therefore PA = B \Rightarrow P = BA^{-1} = \frac{1}{2} \begin{bmatrix} 3 & 9 & 1 & 3 \end{bmatrix} \begin{bmatrix} -1 & 1 & -2 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -21 & 12 & -7 & 4 \end{bmatrix}$ .  $\therefore AQ = B \Rightarrow Q = A^{-1}B = \frac{1}{2} \begin{bmatrix} -1 & 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 9 & 1 & 3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -2 & -6 & -5 & -15 \end{bmatrix}$ .  $\therefore \text{tr}(2(P+Q)) = |(-21) + (4) + (-2) + (-15)| = 34$ .